

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN PHYSIOGRAPHY

MODULE - 4

PHYSIOGRAPHIC DIVISIONS **THE GREAT PLAINS OF INDIA**



2 THE GREAT PLAINS OF NORTH INDIA



- 1 To the south of the Himalayas and to the north of the peninsula lies the Great Plain of North India.
- 2 It is an **aggradational plain** formed by the **depositional work** of the three river systems i.e. the Indus, the Ganga, and the Brahmaputra.
- 3 This plain is also known by the term Indo-Gangetic-Brahmaputra plain.
- 4 This is the largest alluvial tract of the world extending for a length of 3,200 km from the mouth of Indus to the mouth of Ganga.

THE GREAT PLAIN OF NORTH INDIA



2 THE GREAT PLAINS OF NORTH INDIA

5 Of the total length, the Indian sector alone accounts for 2,400 km in length, its average width varies from 150 to 300 kms.

6 It is **widest in the west** where it stretches for about 500 km and its width decreases in the east.

7 The maximum depth of alluvium deposits varies between 1,000-2,000m.

8 The northern boundary of this plain is well defined by the foot hills of the Shivaliks.

9 But, its **southern boundary is wavy irregular line** along the northern edge of the peninsular India.

10 River flowing through this plain, especially those originating in the Himalayas have deposited a thick layer of alluvium throughout the length and the breadth of the plain.

2

GEOMORPHOLOGY OF THE PLAINS



- 1 These plains have characteristic features of mature stage of fluvial landforms such as sand bars, meanders, oxbow lakes and braided channels.
- 2 The Brahmaputra plains are known for their riverine islands and sand bars.
- 3 Most of these areas are subjected to periodic floods and shifting river courses forming braided streams.
- 4 The mouths of these mighty rivers also form some of the largest deltas of the world, for example, the famous Sunderbans delta.

2

GEOMORPHOLOGY OF THE PLAINS



5 The state of Haryana and Delhi form a water divide between the Indus and the Ganga river systems.

6 As opposed to this, the Brahmaputra river flows from the northeast to the southwest direction before it takes an almost 90° southward turn at Dhubri before it enters into Bangladesh.

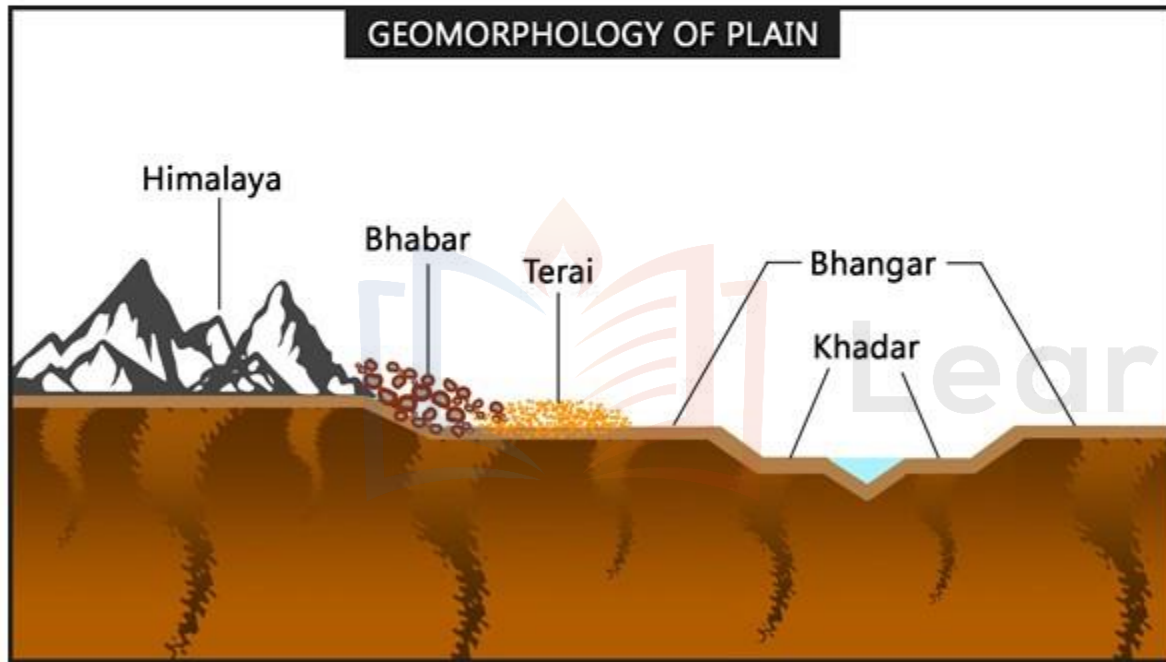
7 These river valley plains have a fertile alluvial soil cover which supports a variety of crops like wheat, rice, sugarcane and jute, and hence, supports a large population.



GEOMORPHOLOGICAL DIVISION VS GEOGRAPHICAL/REGIONAL DIVISION

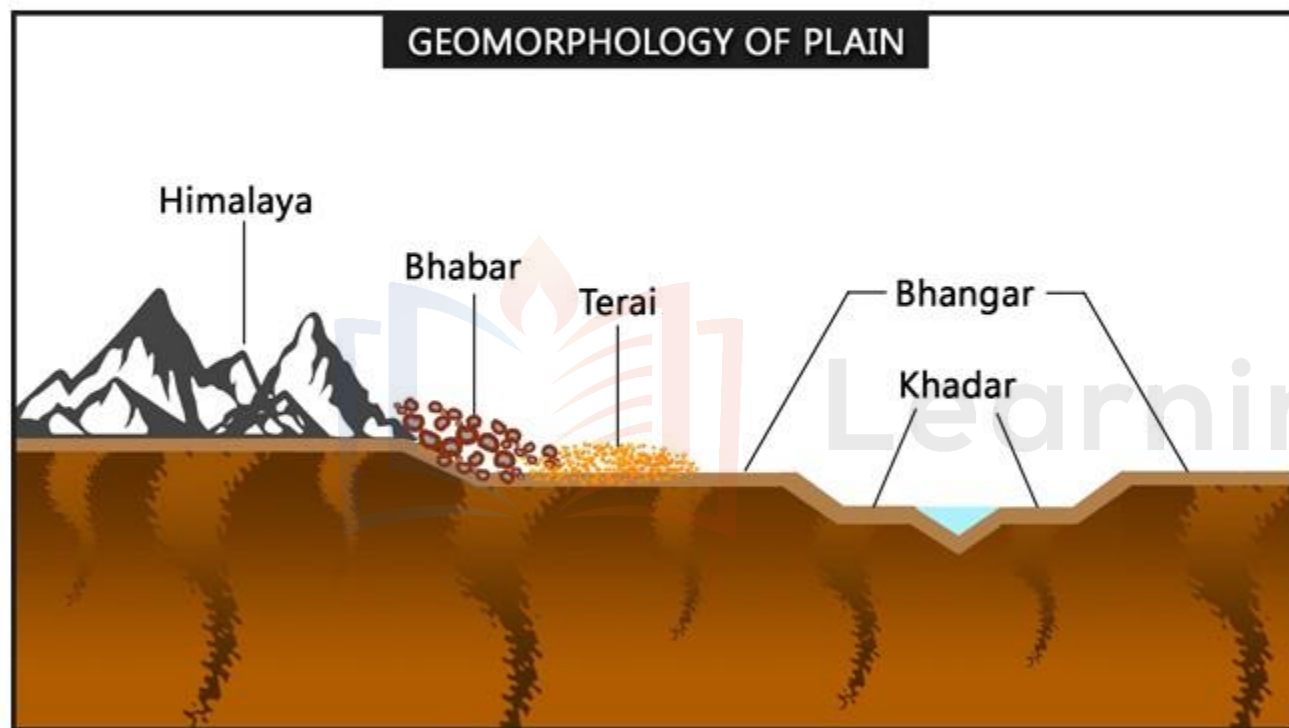
WHAT IS THE DIFFERENCE BETWEEN THE TWO?

GEOMORPHOLOGY OF PLAIN



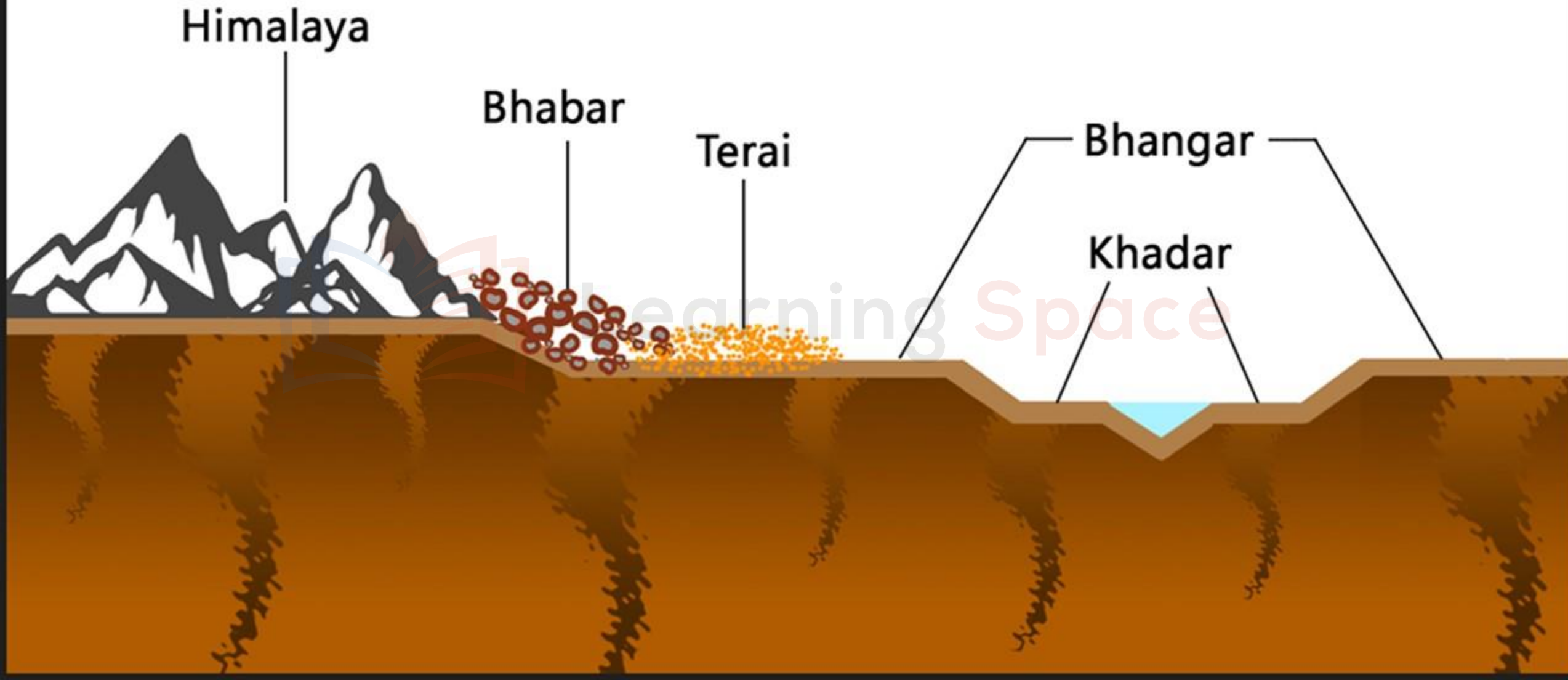
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GEOMORPHOLOGY OF THE PLAINS



- 1 There is a tendency among geographers to treat the plains of north India as a monotonous flat and featureless plain.
- 2 But, it has its own geomorphological varieties which have their own significance.
- 3 The following geomorphological features may be noticed depending upon the relief and soil structure.

GEOMORPHOLOGY OF PLAIN



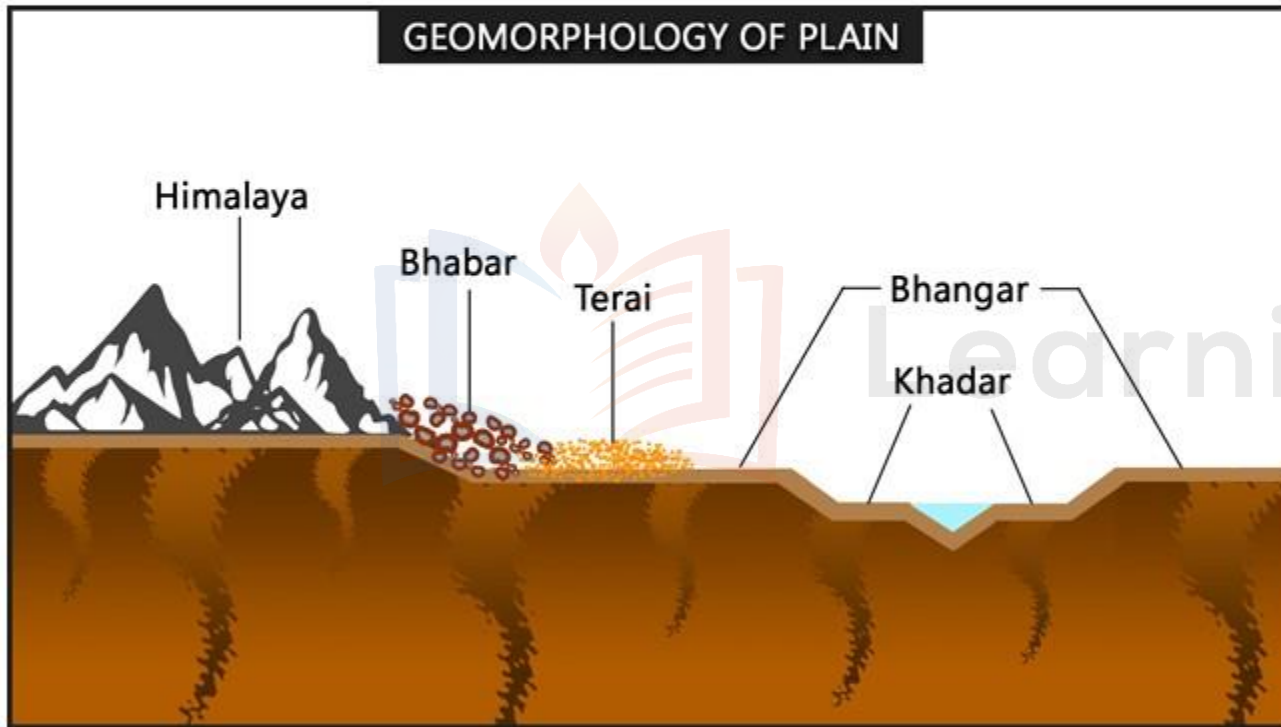
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GEOMORPHOLOGY OF THE PLAINS

1

BHABAR

GEOMORPHOLOGY OF PLAIN



1

The Bhabar is a narrow built belt about 8 to 16 km wide running in east-west direction along the foot hills of Shivaliks with the remarkable continuity from the Indus to the Tista.

2

Rivers descending from the Himalayas deposit their load along the foothills in the form of alluvial fans which form the bhabhar belt.

3

These fans, **consisting of gravel and unassorted sediments**, have merged together to build up the bhabar belt which forms the northern-most component of the Great Plain.

2

GEOMORPHOLOGY OF THE PLAINS

1

BHABAR



Bhabar belt

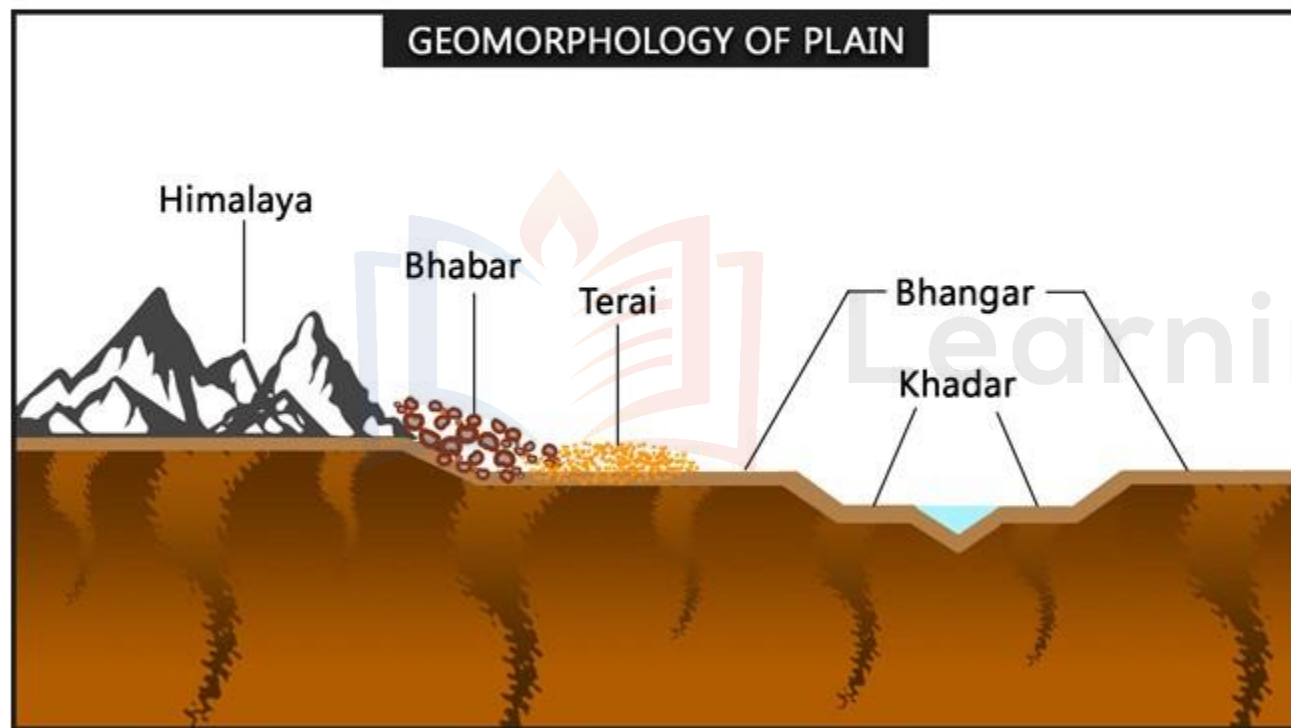
- 4 The porosity of the pebble studded rock beds is so high that most of the streams sink and flow underground.
- 5 Therefore, the area is marked by dry river courses except in the rainy season.
- 6 The Bhabar belt is comparatively narrow in the east and extensive in the western and the north western hilly region.
- 7 The area is not suitable for the agriculture and only the big trees with large roots thrive in this belt.

2

GEOMORPHOLOGY OF THE PLAINS

2

TERAI



1

The Terai is a 15-30 km wide marshy tract in the south of the Bhabar running parallel to it.

2

It is marked by the re-emergence of the underground streams of the Bhabar belt.

3

The re-emerged water converts large areas along the rivers into ill-drained marshy lands of excessive dampness covered with thick forest giving shelter to a variety of wild life.

2

GEOMORPHOLOGY OF THE PLAINS

2

TERAI



4 The Terai is more marked in the eastern part than in the west because the eastern parts of the plain receive higher amounts of rainfall than their western counter parts.

5 Most of the Terai land especially in Punjab, Uttar Pradesh and Uttaranchal has been reclaimed and turned into agricultural land which gives good crops of sugar cane, rice and wheat.

2

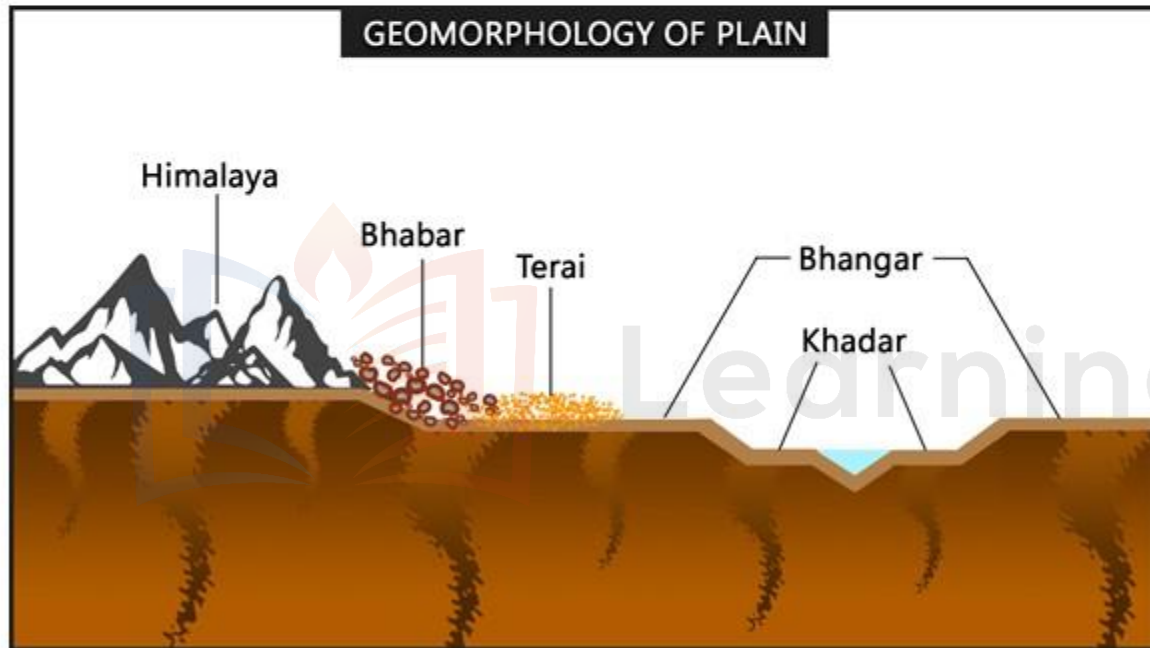
GEOMORPHOLOGY OF THE PLAINS

3

BHANGAR

1

This is composed of **old alluvium** of the **Middle Pleistocene** age and forms the alluvial terrace above the level of flood plains.



2

It is often impregnated with **calcareous concretions** known as **kankar**.

3

It mostly occupies the Pleistocene terrace dating back through Middle and Upper Pleistocene periods.

4

It also contains **remains of animals** like **Rhinoceros**, **Hippopotamus**, **Elephants** etc.

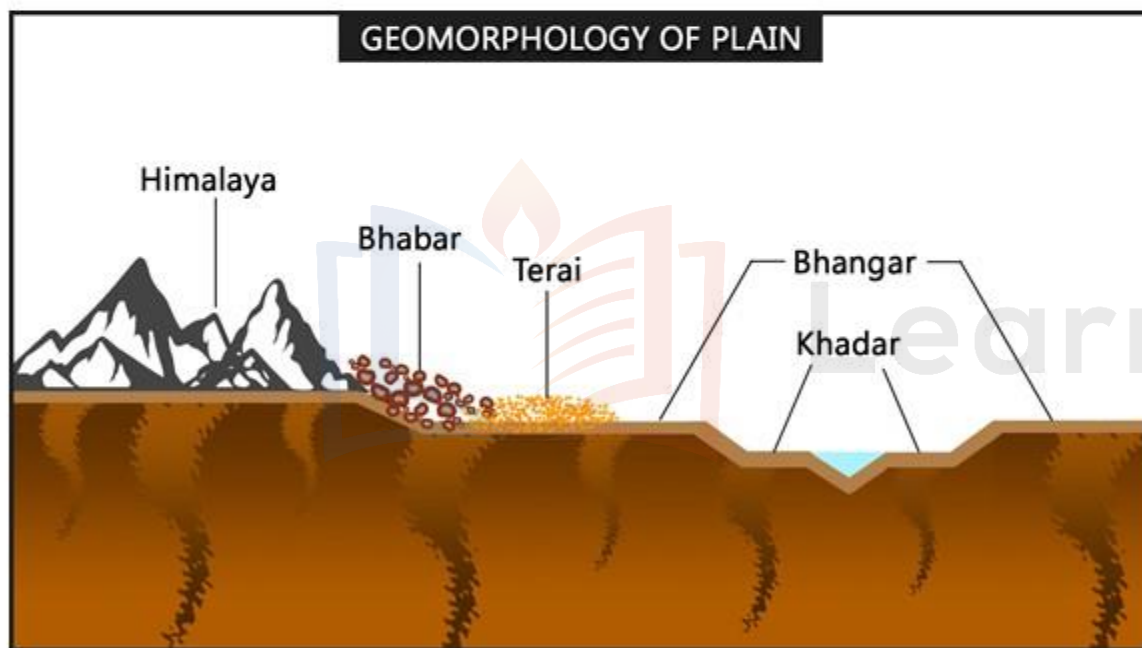
POPULAR QUERIES WITH BHANGAR CONCEPT

- Madam, how the bhangar soil can be clayey as well as full of kankar nodules? i mean clayey itself means the particles are finely packed.
- Mam, alluvial soil are loamy soil so how bhabhar, bhangar can be the eg. of this soil?
- Mam, in the video, differences between kadhar and bhangar...you explained that bhangar is coarser in texture in the second point and also the clayey composition in the fifth point, which is quite contrast and opposite? plzz explain?

2 GEOMORPHOLOGY OF THE PLAINS

4 KHADAR

- 1 This is composed of newer alluviums and forms the flood plains along the river banks.



- 2 A newer layer of alluvium is deposited by river flood almost every year.
- 3 These deposits are normally confined to the vicinity of the present river channels.
- 4 The clays have lesser kankar, and the soil is finer in texture.
- 5 The khadar imperceptibly merges into the deltaic and other accumulations of prehistoric times.

2

GEOMORPHOLOGY OF THE PLAINS

KHADAR (NEW)	BHANGAR (OLD)
1. The area in northern plains that lies below flood levels is known as Khadar which is new alluvium soil. Khadar is younger alluvium as flood waters deposit new layers every year.	1. Largest part of Northern Plains in India is made up of deposits of old alluvium soils known as Bhangar.
2. These soils are finer in texture	2. These are coarser in texture
3. Khadar is very fertile and ideal for intensive cultivation	3. These soils are less fertile as these are found in the higher reaches, about 30m above the flood level and present a terrace like structure.
4. Khadar is less calcareous/ carbonaceous i.e. they are less kankary.	4. Bhangar is full of kankar and calcareous structures
5. They are pale brown in colour and composed mostly of loams & also sandy clays.	5. They are generally dark in colour and of a more clayey composition.

2

GEOMORPHOLOGY OF THE PLAINS

5

REH

- 1 This comprises of barren saline efflorescence of **drier areas** in **Uttar Pradesh and Haryana**.
- 2 Reh areas have spread in recent times with indiscriminate irrigation.

6

BHUR

- 1 Bhur denotes an elevated piece of land situated along the banks of the Ganga river especially in the **upper Ganga-Yamuna Doab**.
- 2 This has been formed due to accumulation of wind-blown sands during the hot dry months of the year.

WHY DO THE AREAS AROUND PUNJAB SUFFER FROM THE PROBLEM OF REH?

- Excessive irrigation, by means of canal irrigation, and fertilization of fields done with the motto of increasing productivity is one important reason for degradation of lands.
- The canal water soaks into the ground and leads to the problem of water-logging along the canal route.
- Excessive waterlogging issues in the fields causes an intensification of capillary action which brings alkaline salts to the surface and makes large areas unfit for agriculture.
- Vast areas in Punjab, Haryana and Uttar Pradesh suffer from the problem of 'reh' caused by canal irrigation.

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